

Soc Internship day 8

1.L1 Analyst

L1 Analyst, is the **first line of defense** against cyber threats. They monitor security systems, identify suspicious activity, and escalate serious incidents to senior analysts

responsibilities:

Monitor alerts from **SIEM** tools (like Splunk, Sentinel, QRadar)

- Check logs and security events
- Perform initial alert triage (real threat vs false positive)
- Investigate suspicious IPs, malware alerts, phishing attempts
- Create incident tickets and documentation
- Escalate confirmed threats to L2 analysts
- Follow security playbooks and SOPs (Standard Operating Procedures)

L2 Analyst

handles deeper security investigations after incidents are escalated from L1 analysts. They perform advanced analysis, threat validation, and incident response.

Responsibilities

Investigate escalated alerts from L1 analysts

- Perform **incident response** and containment
- Analyze malware behavior
- Conduct threat hunting
- Investigate suspicious network traffic and system logs
- Use forensic tools to determine attack scope
- Tune SIEM rules to reduce false positives
- Create detection rules and improve security monitoring
- Coordinate with IT/security teams during incidents

L3 Analyst

L3 analysts handle advanced threats, complex incident response, and security improvements.

Investigate advanced persistent threats (APT) and sophisticated attacks

- Perform advanced **digital forensics** and malware reverse engineering
- Lead major incident response activities
- Conduct proactive **threat hunting**
- Create and improve detection rules (SIEM tuning)
- Develop security playbooks and response procedures
- Analyze attacker tactics using frameworks like **MITRE ATT&CK**
- Mentor L1 and L2 analysts
- Coordinate with security engineers and management during critical incidents
- Perform root cause analysis after security incidents

SOC Escalation Workflow:

Security Event / Alert Generated

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L1 Analyst (Monitoring & Triage)

- **Monitor SIEM alerts**
- **Validate alert**
- **Check logs**
- **Identify false positives**
- **Create incident ticket**

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If simple → Close Alert

If suspicious → Escalate

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L2 Analyst (Investigation & Response)

- **Deep analysis**
- **Correlate logs**
- **Threat hunting**
- **Malware investigation**
- **Incident containment**

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If resolved → Close Incident

If advanced threat → Escalate

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L3 Analyst (Advanced Response)

- **Digital forensics**
- **Malware reverse engineering**
- **Root cause analysis**
- **Advanced threat investigation**
- **Security improvement recommendations**

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SOC Lead / Security Manager

- Decision making
- Business communication
- Compliance reporting
- Long-term remediation

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Post-Incident Review

- Lessons learned
- Update playbooks
- Improve SIEM rules
- Team training

Explain escalation workflow in your own words

An escalation workflow in a SOC (Security Operations Center) is the process of moving a security alert to higher levels of expertise when more analysis is needed. Security events are first checked by L1 analysts for basic validation. If the issue is suspicious or complex, it moves to L2 analysts for deeper investigation. If it becomes an advanced threat, it is escalated to L3 analysts for expert-level analysis, forensics, and response. This workflow ensures threats are handled efficiently and by the right people.

2. Differentiate Triage vs Investigation

Triage	Investigation
Initial assessment of an alert	Deep analysis of a security event
Determines if alert is real or false positive	Determines cause, impact, and scope
Usually performed by L1 analysts	Usually performed by L2 or L3 analysts
Quick decision-making process	Detailed technical examination
Example: Checking login logs after an alert	Example: Finding how attacker entered the system

3. Real-World Scenario and L1, L2, L3 Role Mapping

Scenario: Phishing Email Attack

An employee receives a phishing email pretending to be from the IT department asking to reset their password. The employee clicks the link and enters credentials.

L1 Analyst (Monitoring & Triage)

- SIEM generates suspicious login alert
- Reviews email and login logs
- Validates suspicious activity
- Creates incident ticket

- Escalates to L2

L2 Analyst (Investigation & Response)

- Investigates affected account
- Checks if attacker accessed systems
- Resets compromised password
- Blocks phishing domain
- Searches for similar emails

L3 Analyst (Advanced Response)

- Performs forensic analysis
- Identifies attacker techniques
- Updates email security controls
- Creates new SIEM detection rules
- Improves security awareness training

Outcome:

- ✓ Compromised account secured
- ✓ Phishing campaign blocked
- ✓ Security monitoring improved

This shows how **L1** → **L2** → **L3** work together to detect, investigate, and stop cyber threats.